

EE 671: VLSI Design

PROJECT 1



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Contents

1	Cell: invx1	3
1.1	Circuit Diagram	3
1.2	Transistor Dimensions	3
1.3	Layout	3
1.4	Original Netlist	4
1.5	PEX Netlist	4
1.6	Verification	5
1.7	Timing and Power Characterisation	5
1.7.1	Input Pin Capacitances	5
1.7.2	Transition Time Tables	5
1.7.3	Propagation Delay Time Tables	6
1.7.4	Static Power	6
1.7.5	Dynamic Power Tables	6
1.8	Verilog	6
1.8.1	invx1 Verilog Code	6
1.8.2	Testbench	7
1.8.3	Simulation Waveform	7
2	Cell: a21oi	8
2.1	Circuit Diagram	8
2.2	Transistor Dimensions	8
2.3	Layout	8
2.4	Original Netlist	9
2.5	PEX Netlist	9
2.6	Verification	11
2.7	Waveforms	11
2.8	Timing and Power Characterisation	12
2.8.1	Input Pin Characterization	12
2.8.2	Transition Time Characterization	12
2.8.3	Propagation Delay Characterization	13
2.8.4	Static Power	14
2.8.5	Dynamic Power	15
2.9	Verilog	16
2.9.1	a21oi Verilog Code	16
2.9.2	Testbench	16
2.9.3	Simulation Waveform	16
3	Cell: dfxtn	18
3.1	Circuit Diagram	18
3.2	Transistor Dimensions	18
3.3	Layout	18
3.4	Original Netlist	18
3.5	PEX Netlist	19
3.6	Verification	23
3.7	Waveforms	23
3.8	Timing and Power Characterisation	25
3.8.1	Input Pin Capacitances	25

3.8.2	Set-up Time	25
3.8.3	Hold Time	25
3.8.4	Transition Times (Q output, CLK slew = 10 ps)	25
3.8.5	CLK-to-Q Delay Time	26
3.8.6	Static Power (all input combinations)	26
3.8.7	Dynamic Power (CLK slew = 10 ps)	26
3.9	Verilog	27
3.9.1	dfxtn Verilog Code	27
3.9.2	Testbench	27
3.9.3	Simulation Waveform	28
4	Work Distribution	28

1 Cell: invx1

1.1 Circuit Diagram

Figure 1 shows the schematic of the invx1 cell.

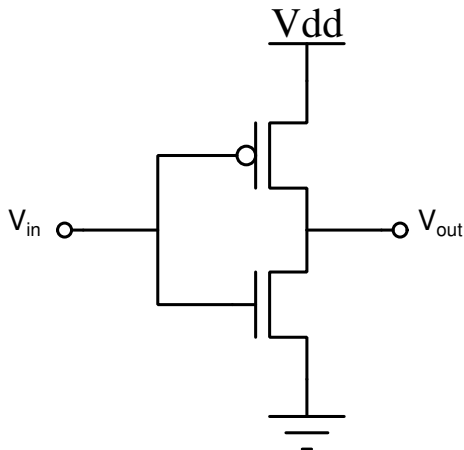


Figure 1: Circuit diagram of the invx1 cell

1.2 Transistor Dimensions

The width and length of the MOSFETs used in the cell are listed in Table 1.

	Width (μm)	Length (μm)
PMOS	1.26	0.15
NMOS	0.42	0.15

Table 1: MOSFET width and length values for the invx1 cell

1.3 Layout

The layout of the invx1 cell is shown in Figure 2. The cell area is measured as:

$$\text{Area} = 2.61\mu \times 2.72\mu = 7.099\mu m^2$$

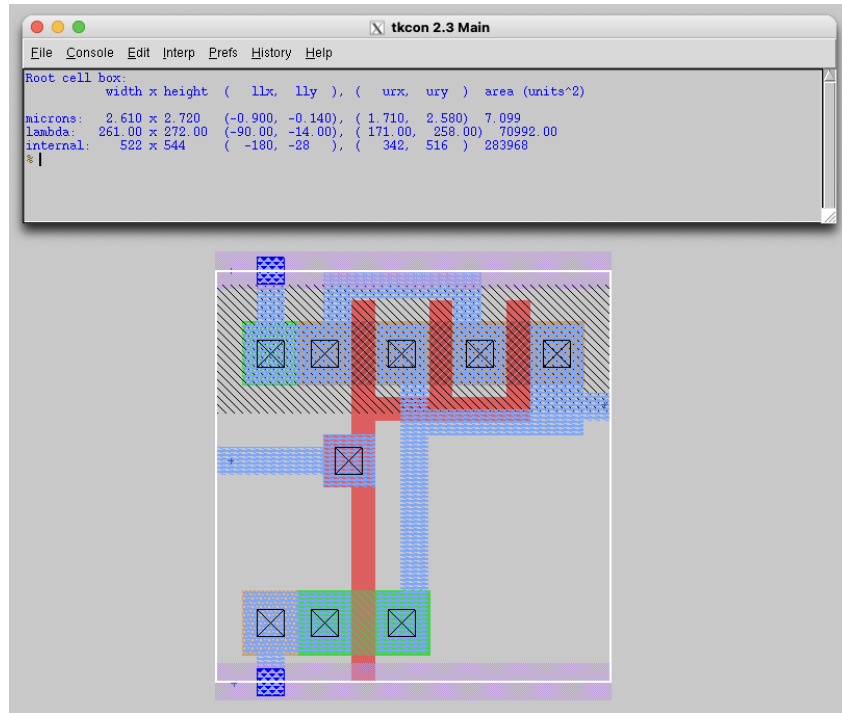


Figure 2: Layout of the invx1 cell

1.4 Original Netlist

The netlist written for the invx1 cell is shown below:

```

1 .subckt invx1_sch in out vdd vss
2 xmp out in vdd vdd sky130_fd_pr__pfet_01v8 l=0.15 w=1.26 as=0.302
   ad=0.302 ps=3.12 pd=3.12
3 xmn out in vss vss sky130_fd_pr__nfet_01v8 l=0.15 w=0.42 as=0.151
   ad=0.151 ps=1.52 pd=1.52
4 .ends

```

1.5 PEX Netlist

The extracted netlist of the cell is shown below:

```

1 * NGSPICE file created from invx1.ext - technology: sky130A
2
3 .subckt invx1 in out vdd vss
4 X0 vdd in out vdd sky130_fd_pr__pfet_01v8 ad=0.1008 pd=1.04 as
   =0.1008 ps=1.04 w=0.42 l=0.15
5 **devattr s=3024,156 d=3024,156
6 X1 out in vdd vdd sky130_fd_pr__pfet_01v8 ad=0.1008 pd=1.04 as
   =0.1008 ps=1.04 w=0.42 l=0.15
7 **devattr s=3024,156 d=6048,312
8 X2 out in vss vss sky130_fd_pr__nfet_01v8 ad=0.1512 pd=1.56 as
   =0.1512 ps=1.56 w=0.42 l=0.15
9 **devattr s=6048,312 d=6048,312
10 X3 out in vdd vdd sky130_fd_pr__pfet_01v8 ad=0.1008 pd=1.04 as
   =0.1008 ps=1.04 w=0.42 l=0.15

```

```

11 **devattr s=6048,312 d=3024,156
12 C0 in out 0.09754f
13 C1 vdd out 0.20135f
14 C2 vdd in 0.19312f
15 C3 out vss 0.2797f
16 C4 in vss 0.46461f
17 C5 vdd vss 0.5609f
18 .ends

```

1.6 Verification

The DRC and LVS results are clean (Figure 3).

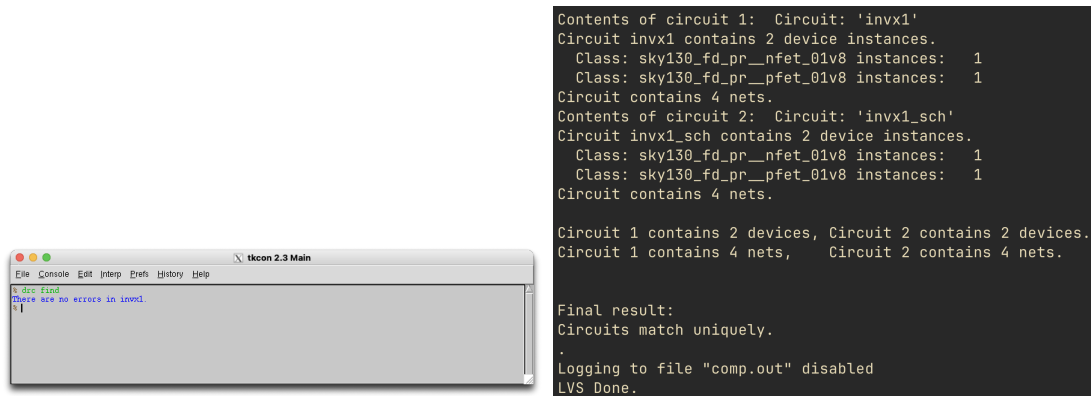


Figure 3: DRC and LVS clean results for the invx1 cell

1.7 Timing and Power Characterisation

1.7.1 Input Pin Capacitances

Input Pin	Rise Cap (fF)	Fall Cap (fF)	Average Cap (fF)
A	2.915	2.915	2.915

Table 2: Input pin capacitance values

1.7.2 Transition Time Tables

	10 ps	100 ps	1000 ps
0.5 fF	0.014	0.023	0.084
10 fF	0.066	0.068	0.173
100 fF	0.577	0.577	0.588

Table 3: Output Rise Transition Time (ns) — Input Slew vs Output Capacitance (Related pin A, others held constant)

	10 ps	100 ps	1000 ps
0.5 fF	0.014	0.021	0.084
10 fF	0.073	0.073	0.174
100 fF	0.642	0.642	0.648

Table 4: Output Fall Transition Time (ns) — Input Slew vs Output Capacitance (Related pin A, others held constant)

1.7.3 Propagation Delay Time Tables

	10 ps	100 ps	1000 ps
0.5 fF	0.014	0.033	0.085
10 fF	0.051	0.075	0.222
100 fF	0.399	0.423	0.658

Table 5: Cell Rise Delay (ns) — Input Slew vs Output Capacitance (Related pin A, others held constant)

	10 ps	100 ps	1000 ps
0.5 fF	0.018	0.034	0.078
10 fF	0.065	0.085	0.216
100 fF	0.501	0.521	0.727

Table 6: Cell Fall Delay (ns) — Input Slew vs Output Capacitance (Related pin A, others held constant)

1.7.4 Static Power

Condition (A)	Power (nW)
0	0.44
1	0.108

Table 7: Static power dissipation under different input conditions

1.7.5 Dynamic Power Tables

	10 ps	100 ps	1000 ps
0.5 fF	3857	3879	6690
10 fF	19454	19304	20977
100 fF	150969	148418	158437

Table 8: Dynamic Rise Power (nW) — Input Slew vs Output Capacitance (Related pin A, others held constant)

1.8 Verilog

1.8.1 invx1 Verilog Code

The Verilog code for the invx1 cell used in simulations is shown below:

Listing 1: invx1 Verilog code

```

1 module inverter (
2     input wire a,
3     output wire y
4 );
5     assign y = ~a;
6 endmodule

```

1.8.2 Testbench

The Verilog testbench used to verify the invx1 functionality is shown below:

Listing 2: invx1 Testbench code

```

1 `timescale 1ns/1ps
2
3 module tb;
4     reg a;
5     wire y;
6
7     inverter uut (
8         .a(a),
9         .y(y)
10    );
11
12    initial begin
13        $dumpfile("dump.vcd");
14        $dumpvars(0, tb);
15        $monitor("Time=%0t | a=%b | y=%b", $time, a, y);
16
17        a = 0; #10;
18        a = 1; #10;
19        a = 0; #10;
20
21        $finish;
22    end
23 endmodule

```

1.8.3 Simulation Waveform

The waveform from the Verilog simulation showing the input, output, and clock signals is shown in Figure 4.

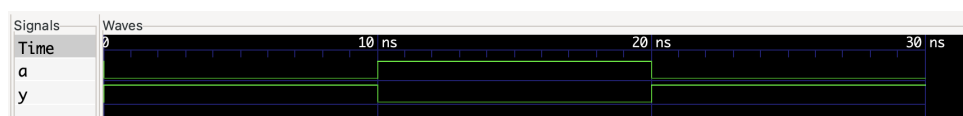


Figure 4: Verilog simulation waveform of the invx1 cell

2 Cell: a21oi

2.1 Circuit Diagram

Figure 5 shows the schematic of the a21oi cell.

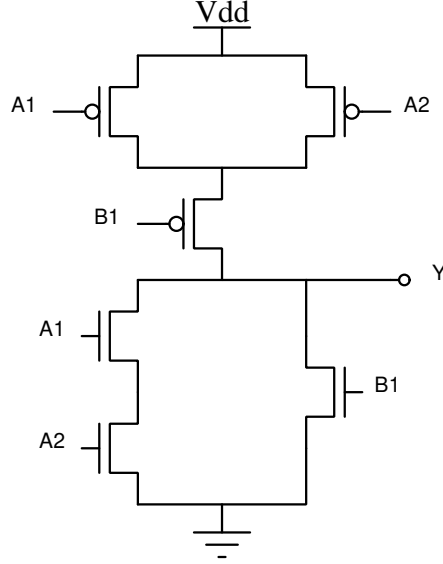


Figure 5: Circuit diagram of the a21oi cell

2.2 Transistor Dimensions

The width and length of the MOSFETs used in the cell are listed in Table 9.

	Width (μm)	Length (μm)
PMOS _{A1}	3	0.15
NMOS _{A1}	0.9	0.15
PMOS _{A2}	3	0.15
NMOS _{A2}	0.9	0.15
PMOS _{B1}	3	0.15
NMOS _{B1}	0.45	0.15

Table 9: MOSFET width and length values for the a21oi cell

2.3 Layout

The layout of the a21oi cell is shown in Figure 6. The cell area is measured as:

$$\text{Area} = 9.93\mu \times 2.72\mu = 27.01\mu m^2$$

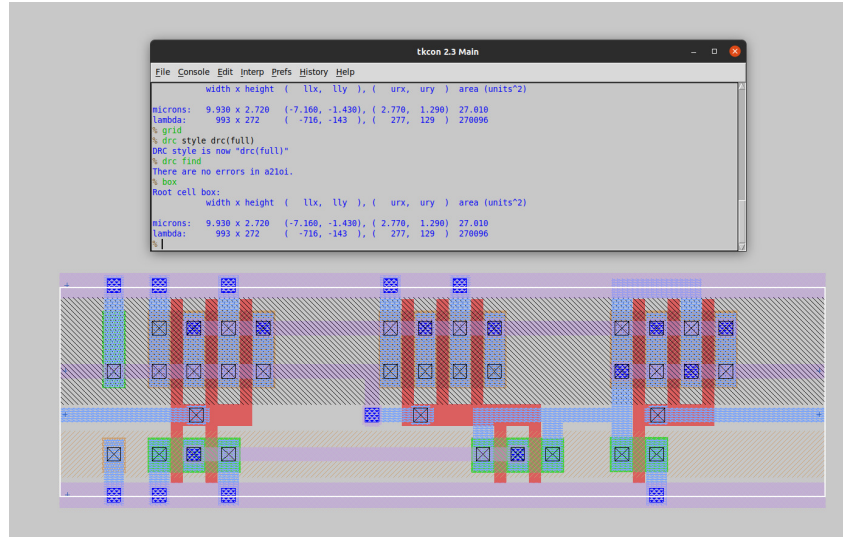


Figure 6: Layout of the a2loi cell

2.4 Original Netlist

The netlist written for the a2loi cell is shown below:

```

1  .subckt a2loi a1 a2 b1 vdd vss y
2
3  xpm_a1 na a1 vdd vdd sky130_fd_pr__pfet_01v8 l=0.15 w=3 as=0.9 ad
   =0.9 ps=6.6 pd=6.6
4  xpm_a2 na a2 vdd vdd sky130_fd_pr__pfet_01v8 l=0.15 w=3 as=0.9 ad
   =0.9 ps=6.6 pd=6.6
5  xpm_b1 y b1 na vdd sky130_fd_pr__pfet_01v8 l=0.15 w=3 as=0.9 ad
   =0.9 ps=6.6 pd=6.6
6
7
8  xnm_a1 y a1 nb vss sky130_fd_pr__nfet_01v8 l=0.15 w=0.9 as=0.27
   ad=0.27 ps=2.4 pd=2.4
9  xnm_a2 nb a2 vss vss sky130_fd_pr__nfet_01v8 l=0.15 w=0.9 as=0.27
   ad=0.27 ps=2.4 pd=2.4
10 xnm_b1 y b1 vss vss sky130_fd_pr__nfet_01v8 l=0.15 w=0.45 as
    =0.135 ad=0.135 ps=1.5 pd=1.5
11
12 .ends

```

2.5 PEX Netlist

The extracted netlist of the cell is shown below:

```

1  * NGSPICE file created from a2loi.ext - technology: sky130A
2
3  .subckt a2loi a1 a2 b1 vdd vss y
4  X0 a_n1114_n4# a1 vdd vdd sky130_fd_pr__pfet_01v8 ad=0.2 pd
   =1.73333 as=0.2 ps=1.73333 w=1 l=0.15
5  **devattr s=6000,260 d=12000,520

```

```

6 X1 a_n1114_n4# a2 vdd vdd sky130_fd_pr__pfet_01v8 ad=0.2 pd
   =1.73333 as=0.2 ps=1.73333 w=1 l=0.15
7 **devattr s=12000,520 d=6000,260
8 X2 a_n1114_n4# b1 y vdd sky130_fd_pr__pfet_01v8 ad=0.2 pd=1.73333
   as=0.2 ps=1.73333 w=1 l=0.15
9 **devattr s=6000,260 d=12000,520
10 X3 a_n1114_n4# a1 vdd vdd sky130_fd_pr__pfet_01v8 ad=0.2 pd
   =1.73333 as=0.2 ps=1.73333 w=1 l=0.15
11 **devattr s=12000,520 d=6000,260
12 X4 a_n1114_n4# b1 y vdd sky130_fd_pr__pfet_01v8 ad=0.2 pd=1.73333
   as=0.2 ps=1.73333 w=1 l=0.15
13 **devattr s=12000,520 d=6000,260
14 X5 vss b1 y vss sky130_fd_pr__nfet_01v8 ad=0.135 pd=1.5 as=0.135
   ps=1.5 w=0.45 l=0.15
15 **devattr s=5400,300 d=5400,300
16 X6 vdd a2 a_n1114_n4# vdd sky130_fd_pr__pfet_01v8 ad=0.2 pd
   =1.73333 as=0.2 ps=1.73333 w=1 l=0.15
17 **devattr s=6000,260 d=6000,260
18 X7 a_n1114_n4# a2 vdd vdd sky130_fd_pr__pfet_01v8 ad=0.2 pd
   =1.73333 as=0.2 ps=1.73333 w=1 l=0.15
19 **devattr s=6000,260 d=12000,520
20 X8 y b1 a_n1114_n4# vdd sky130_fd_pr__pfet_01v8 ad=0.2 pd=1.73333
   as=0.2 ps=1.73333 w=1 l=0.15
21 **devattr s=6000,260 d=6000,260
22 X9 vdd a1 a_n1114_n4# vdd sky130_fd_pr__pfet_01v8 ad=0.2 pd
   =1.73333 as=0.2 ps=1.73333 w=1 l=0.15
23 **devattr s=6000,260 d=6000,260
24 X10 vss a2 a_n1114_n224# vss sky130_fd_pr__nfet_01v8 ad=0.135 pd
   =1.5 as=0.0675 ps=0.75 w=0.45 l=0.15
25 **devattr s=2700,150 d=5400,300
26 X11 a_n1114_n224# a2 vss vss sky130_fd_pr__nfet_01v8 ad=0.0675 pd
   =0.75 as=0.135 ps=1.5 w=0.45 l=0.15
27 **devattr s=5400,300 d=2700,150
28 X12 y a1 a_n1114_n224# vss sky130_fd_pr__nfet_01v8 ad=0.135 pd
   =1.5 as=0.0675 ps=0.75 w=0.45 l=0.15
29 **devattr s=2700,150 d=5400,300
30 X13 a_n1114_n224# a1 y vss sky130_fd_pr__nfet_01v8 ad=0.0675 pd
   =0.75 as=0.135 ps=1.5 w=0.45 l=0.15
31 **devattr s=5400,300 d=2700,150
32 C0 b1 y 0.15376f
33 C1 a1 y 0.08047f
34 C2 a2 y 0
35 C3 a1 b1 0.01136f
36 C4 vdd y 0.22587f
37 C5 a2 a1 0.0525f
38 C6 vdd b1 0.13852f
39 C7 vdd a1 0.40834f
40 C8 a_n1114_n224# y 0.11833f
41 C9 vdd a2 0.21072f
42 C10 a_n1114_n224# b1 0
43 C11 a_n1114_n4# y 0.50282f

```

```

44 C12 a_n1114_n224# a1 0.14769f
45 C13 a_n1114_n4# b1 0.07687f
46 C14 a_n1114_n224# a2 0.04479f
47 C15 a_n1114_n4# a1 0.29918f
48 C16 a_n1114_n224# vdd 0.01848f
49 C17 a_n1114_n4# a2 0.05658f
50 C18 a_n1114_n4# vdd 1.28977f
51 C19 a_n1114_n4# a_n1114_n224# 0.05679f
52 C20 y vss 0.41345f
53 C21 b1 vss 0.43351f
54 C22 a1 vss 0.50924f
55 C23 a2 vss 0.45217f
56 C24 vdd vss 2.28043f
57 C25 a_n1114_n224# vss 0.56133f
58 C26 a_n1114_n4# vss 0.11157f
59 .ends

```

2.6 Verification

The DRC and LVS results are clean (Figure 7).

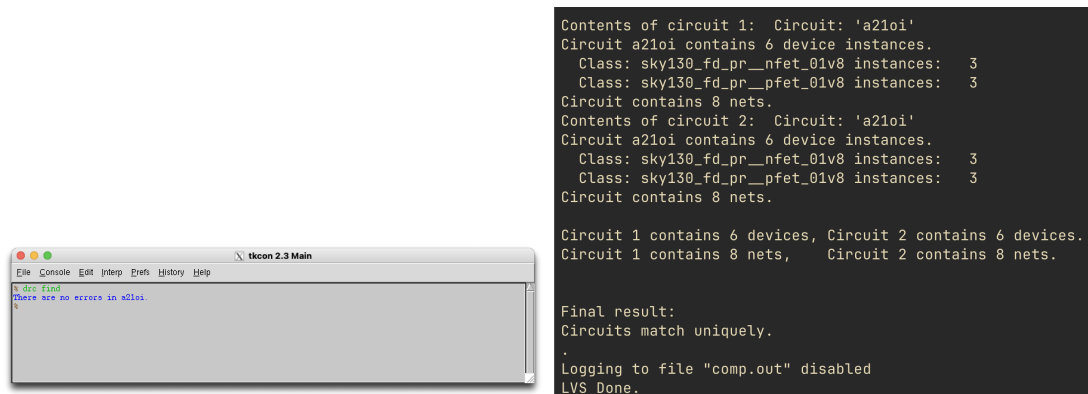


Figure 7: DRC and LVS clean results for the a21oi cell

2.7 Waveforms

Shown below is one representative waveform ($a_2 = 1$, $b_1 = 0$, a_1 pulsing) used to calculate t_{rise} , t_{fall} , and $t_{propagation}$.

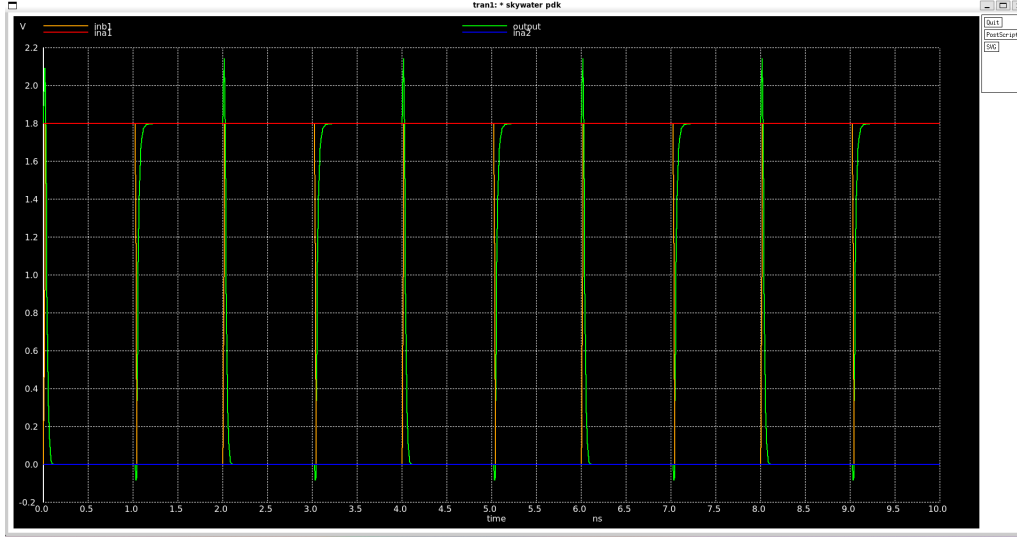


Figure 8: Simulation waveform for $a_2 = 1$, $b_1 = 0$, a_1 pulsing

2.8 Timing and Power Characterisation

2.8.1 Input Pin Characterization

Input Pins	Rise Cap (fF)	Fall Cap (fF)	Average Cap (fF)
A	3.86	3.95	3.91
B	4.42	4.40	4.41
C	4.175	4.173	4.174

Table 10: Input pin capacitances.

2.8.2 Transition Time Characterization

Output Cap	10 ps	100 ps	1000 ps
0.5 fF	0.026	0.028	0.026
10 fF	0.072	0.072	0.076
100 fF	0.515	0.515	0.643

Table 11: Output rise transition times (ns) with respect to related pin A.

Output Cap	10 ps	100 ps	1000 ps
0.5 fF	0.03298	0.03392	0.03392
10 fF	0.07981	0.08668	0.1449
100 fF	0.52283	0.51057	0.49776

Table 12: Output rise transition times (ns) with respect to related pin B.

Output Cap	10 ps	100 ps	1000 ps
0.5 fF	0.03127	0.03495	0.03189
10 fF	0.05173	0.0539	0.05467
100 fF	0.381	0.381	0.37633

Table 13: Output rise transition times (ns) with respect to related pin C.

Output Cap	10 ps	100 ps	1000 ps
0.5 fF	0.04111	0.04046	0.10865
10 fF	0.08666	0.08636	0.17017
100 fF	0.51058	0.51060	0.52165

Table 14: Output fall transition times (ns) with respect to related pin A.

Output Cap	10 ps	100 ps	1000 ps
0.5 fF	0.04144	0.04079	0.04079
10 fF	0.08681	0.08668	0.1449
100 fF	0.5106	0.51057	0.52132

Table 15: Output fall transition times (ns) with respect to related pin B.

Output Cap	10 ps	100 ps	1000 ps
0.5 fF	0.03747	0.0409	0.14135
10 fF	0.0795	0.0797	0.177
100 fF	0.6122	0.6122	0.64343

Table 16: Output fall transition times (ns) with respect to related pin C.

2.8.3 Propagation Delay Characterization

Output Cap	10 ps	100 ps	1000 ps
0.5 fF	1.04698	1.15646	1.5353
10 fF	1.0809	1.19140	1.5697
100 fF	1.36655	1.4856	1.53388

Table 17: Cell rise delay (ns) with respect to related pin A.

Output Cap	10 ps	100 ps	1000 ps
0.5 fF	1.04043	1.13987	1.54399
10 fF	1.03033	1.16722	1.54833
100 fF	1.26634	1.36587	1.53499

Table 18: Cell rise delay (ns) with respect to related pin B.

Output Cap	10 ps	100 ps	1000 ps
0.5 fF	1.0376	1.1457	1.5267
10 fF	1.0506	1.160	1.54035
100 fF	1.2385	1.3577	1.58522

Table 19: Cell rise delay (ns) with respect to related pin C.

Output Cap	10 ps	100 ps	1000 ps
0.5 fF	0.05031	0.06627	0.16822
10 fF	0.0884	0.10503	0.24213
100 fF	0.41377	0.43106	0.36587

Table 20: Cell fall delay (ns) with respect to related pin A.

Output Cap	10 ps	100 ps	1000 ps
0.5 fF	0.04055	0.05032	0.14377
10 fF	0.08076	0.10054	0.23876
100 fF	0.45054	0.49322	0.40053

Table 21: Cell fall delay (ns) with respect to related pin B.

Output Cap	10 ps	100 ps	1000 ps
0.5 fF	0.0332	0.04944	0.12368
10 fF	0.07736	0.09445	0.23178
100 fF	0.48637	0.50775	0.65499

Table 22: Cell fall delay (ns) with respect to related pin C.

2.8.4 Static Power

Condition (ABC)	Power (nW)
000	131
001	1.72
010	54.4
011	65.8
100	4.9
101	1.72
110	0.05
111	0.05

Table 23: Static power consumption under different input conditions.

2.8.5 Dynamic Power

Output Cap	10 ps	100 ps	1000 ps
0.5 fF	12171	11763	12414
10 fF	27749	27439	27504
100 fF	164833	163974	153631

Table 24: Dynamic rise power (nW) with respect to related pin A.

Output Cap	10 ps	100 ps	1000 ps
0.5 fF	14073	13812	14277
10 fF	29483	29306	29461
100 fF	166182	165269	154771

Table 25: Dynamic rise power (nW) with respect to related pin B.

Output Cap	10 ps	100 ps	1000 ps
0.5 fF	8356	8149	10854
10 fF	23911	23723	25322
100 fF	168004	167765	163504

Table 26: Dynamic rise power (nW) with respect to related pin C.

Output Cap	10 ps	100 ps	1000 ps
0.5 fF	2800	2050	2472
10 fF	2856	2268	2469
100 fF	1868	1485	1099

Table 27: Dynamic fall power (nW) with respect to related pin A.

Output Cap	10 ps	100 ps	1000 ps
0.5 fF	2870	2162	2374
10 fF	2832	2285	2153
100 fF	1776	1402	982

Table 28: Dynamic fall power (nW) with respect to related pin B.

Output Cap	10 ps	100 ps	1000 ps
0.5 fF	1809	2625	945
10 fF	1486	2149	2012
100 fF	1382	1393	2194

Table 29: Dynamic fall power (nW) with respect to related pin C.

2.9 Verilog

2.9.1 a2loi Verilog Code

The Verilog code for the a2loi cell used in simulations is shown below:

Listing 3: a2loi Verilog code

```
1 module logic_gate (  
2     input a1,  
3     input a2,  
4     input b1,  
5     output y  
6 );  
7     assign y = ~((a1 & a2) | b1);  
8 endmodule
```

2.9.2 Testbench

The Verilog testbench used to verify the a2loi functionality is shown below:

Listing 4: a2loi Testbench code

```
1 `timescale 1ns/1ps  
2  
3 module tb;  
4     reg a1, a2, b1;  
5     wire y;  
6  
7     logic_gate dut(.a1(a1), .a2(a2), .b1(b1), .y(y));  
8  
9     initial begin  
10         $dumpfile("dump.vcd");  
11         $dumpvars(0, tb);  
12  
13         a1=0; a2=0; b1=0; #10;  
14         a1=0; a2=0; b1=1; #10;  
15         a1=0; a2=1; b1=0; #10;  
16         a1=0; a2=1; b1=1; #10;  
17         a1=1; a2=0; b1=0; #10;  
18         a1=1; a2=0; b1=1; #10;  
19         a1=1; a2=1; b1=0; #10;  
20         a1=1; a2=1; b1=1; #10;  
21  
22         $finish;  
23     end  
24 endmodule
```

2.9.3 Simulation Waveform

The waveform from the Verilog simulation showing the input, output, and clock signals is shown in Figure 9.

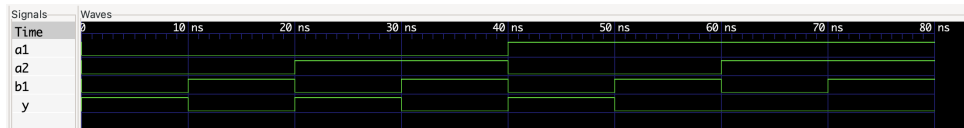


Figure 9: Verilog simulation waveform of the a21oi cell

3 Cell: dfxtn

3.1 Circuit Diagram

Figure 10 shows the schematic of the dfxtn cell.

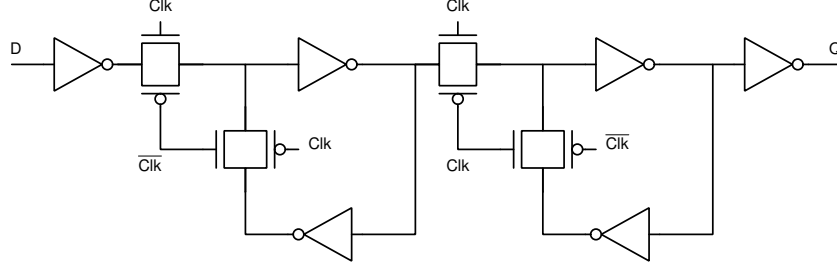


Figure 10: Circuit diagram of the dfxtn cell

3.2 Transistor Dimensions

The width and length of the MOSFETs used in the cell are listed in Table 30.

	Width (μm)	Length (μm)
PMOS	1.26	0.15
NMOS	0.42	0.15

Table 30: MOSFET width and length values for the dfxtn cell

3.3 Layout

The layout of the dfxtn cell is shown in Figure 11. The cell area is measured as:

$$\text{Area} = 31.95\mu \times 2.72\mu = 86.904\mu\text{m}^2$$

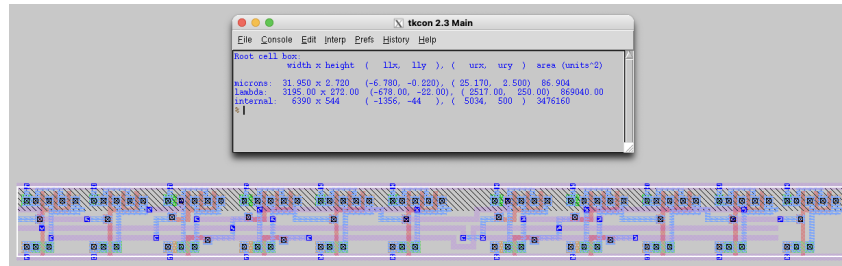


Figure 11: Layout of the dfxtn cell

3.4 Original Netlist

The netlist written for the dfxtn cell is shown below:

```
1 .include invx1_schematic.spice
2 .include tg_schematic.spice
3 .include latch_schematic.spice
```

```

4
5 .subckt dfxtn d q c vdd vss
6
7 Xinvcclk c cb vdd vss invx1_sch
8
9 Xinp d db vdd vss invx1_sch
10
11 Xmaster db y c cb vdd vss latch
12 Xslave y qb cb c vdd vss latch
13
14 Xout qb q vdd vss invx1_sch
15
16 .ends

```

3.5 PEX Netlist

The extracted netlist of the cell is shown below:

```

1 * NGSPICE file created from dfxtn.ext - technology: sky130A
2
3 .subckt tg in out en enb vdd vss
4 X0 in enb out vdd sky130_fd_pr__pfet_01v8 ad=0.1008 pd=1.04 as
   =0.1008 ps=1.04 w=0.42 l=0.15
5 **devattr s=3024,156 d=3024,156
6 X1 out enb in vdd sky130_fd_pr__pfet_01v8 ad=0.1008 pd=1.04 as
   =0.1008 ps=1.04 w=0.42 l=0.15
7 **devattr s=3024,156 d=6048,312
8 X2 out en in vss sky130_fd_pr__nfet_01v8 ad=0.1512 pd=1.56 as
   =0.1512 ps=1.56 w=0.42 l=0.15
9 **devattr s=6048,312 d=6048,312
10 X3 out enb in vdd sky130_fd_pr__pfet_01v8 ad=0.1008 pd=1.04 as
   =0.1008 ps=1.04 w=0.42 l=0.15
11 **devattr s=6048,312 d=3024,156
12 C0 out en 0.07151f
13 C1 in en 0.00907f
14 C2 in out 0.26956f
15 C3 enb en 0.01993f
16 C4 vdd en 0.00403f
17 C5 enb out 0.05681f
18 C6 vdd out 0.03231f
19 C7 enb in 0.08671f
20 C8 vdd in 0.12663f
21 C9 vdd enb 0.15588f
22 C10 en vss 0.29863f
23 C11 out vss 0.1594f
24 C12 in vss 0.19345f
25 C13 enb vss 0.34859f
26 C14 vdd vss 0.52519f
27 .ends
28
29 .subckt invx1 in out vdd vss

```

```

30 X0 vdd in out vdd sky130_fd_pr__pfet_01v8 ad=0.1008 pd=1.04 as
    =0.1008 ps=1.04 w=0.42 l=0.15
31 **devattr s=3024,156 d=3024,156
32 X1 out in vdd vdd sky130_fd_pr__pfet_01v8 ad=0.1008 pd=1.04 as
    =0.1008 ps=1.04 w=0.42 l=0.15
33 **devattr s=3024,156 d=6048,312
34 X2 out in vss vss sky130_fd_pr__nfet_01v8 ad=0.1512 pd=1.56 as
    =0.1512 ps=1.56 w=0.42 l=0.15
35 **devattr s=6048,312 d=6048,312
36 X3 out in vdd vdd sky130_fd_pr__pfet_01v8 ad=0.1008 pd=1.04 as
    =0.1008 ps=1.04 w=0.42 l=0.15
37 **devattr s=6048,312 d=3024,156
38 C0 in out 0.09754f
39 C1 vdd out 0.20135f
40 C2 vdd in 0.19312f
41 C3 out vss 0.2797f
42 C4 in vss 0.46461f
43 C5 vdd vss 0.5609f
44 .ends
45
46 .subckt latch in en tg_1/out vdd enb out tg_1/in vss
47 Xtg_0 in tg_1/out en enb vdd vss tg
48 Xtg_1 tg_1/in tg_1/out enb en vdd vss tg
49 Xinvx1_0 tg_1/out out vdd vss invx1
50 Xinvx1_1 out tg_1/in vdd vss invx1
51 C0 out tg_1/in 0.24428f
52 C1 enb tg_1/in 0.056f
53 C2 vdd tg_1/in 0.39868f
54 C3 enb out 0.04708f
55 C4 enb vdd 0.00241f
56 C5 vdd out 0.0617f
57 C6 en tg_1/in 0.34398f
58 C7 tg_1/out tg_1/in 0.22086f
59 C8 en out 0.25622f
60 C9 en enb 1.28793f
61 C10 tg_1/out out 0.02423f
62 C11 enb tg_1/out 0.0893f
63 C12 en vdd 0.01201f
64 C13 tg_1/out vdd 0.1387f
65 C14 in tg_1/in 0.00354f
66 C15 en tg_1/out 0.48927f
67 C16 in out 0
68 C17 in enb 0.07412f
69 C18 in vdd 0.1056f
70 C19 in en 0.09005f
71 C20 in tg_1/out 0.01806f
72 C21 tg_1/in vss 0.44879f
73 C22 out vss 0.69348f
74 C23 vdd vss 1.84585f
75 C24 tg_1/out vss 0.69694f
76 C25 enb vss 1.52166f

```

```

77 C26 en vss 0.76901f
78 C27 in vss 0.18603f
79 .ends
80
81 .subckt dfxtn d q c vdd vss
82 Xlatch_0 latch_0/in c latch_0/tg_1/out vdd latch_1/en latch_1/in
    latch_0/tg_1/in vss
83 + latch
84 Xlatch_1 latch_1/in latch_1/en latch_1/tg_1/out vdd c invx1_2/in
    latch_1/tg_1/in vss
85 + latch
86 Xinvx1_0 d latch_0/in vdd vss invx1
87 Xinvx1_1 c latch_1/en vdd vss invx1
88 Xinvx1_2 invx1_2/in q vdd vss invx1
89 C0 invx1_2/in latch_1/tg_1/in 0.0088f
90 C1 vdd latch_1/tg_1/in 0.02553f
91 C2 vdd invx1_2/in 0.01506f
92 C3 c invx1_2/in 0.00124f
93 C4 c vdd 0.07558f
94 C5 latch_1/en latch_1/tg_1/in 0
95 C6 latch_1/en invx1_2/in 0.00151f
96 C7 latch_1/en vdd 0.02357f
97 C8 latch_1/in latch_1/tg_1/in 0
98 C9 c latch_1/tg_1/out 0
99 C10 latch_0/tg_1/in vdd 0.0196f
100 C11 latch_1/in invx1_2/in 0
101 C12 latch_1/in vdd 0.06497f
102 C13 latch_1/en latch_1/tg_1/out 0
103 C14 latch_0/tg_1/in latch_1/tg_1/out 0
104 C15 latch_1/in latch_1/tg_1/out 0.00598f
105 C16 latch_1/en c 0.66205f
106 C17 latch_0/tg_1/in c 0.01285f
107 C18 latch_1/in c 0.0848f
108 C19 d vdd 0.15914f
109 C20 latch_0/tg_1/in latch_1/en 0.00222f
110 C21 latch_1/in latch_1/en 0.04251f
111 C22 latch_0/tg_1/out c 0
112 C23 latch_1/in latch_0/tg_1/in 0.01207f
113 C24 latch_0/tg_1/out latch_1/en 0
114 C25 d c 0.36115f
115 C26 q latch_1/tg_1/in 0.01338f
116 C27 latch_0/in vdd 0.035f
117 C28 latch_0/tg_1/out latch_1/in 0
118 C29 q invx1_2/in 0
119 C30 q vdd 0
120 C31 d latch_1/en 0.09262f
121 C32 d latch_0/tg_1/in 0
122 C33 latch_0/in c 0.05593f
123 C34 q c 0
124 C35 d latch_0/tg_1/out 0.00288f
125 C36 latch_0/in latch_1/en 0.06449f

```

```
126 C37 latch_0/in latch_0/tg_1/in 0
127 C38 q latch_1/en 0
128 C39 latch_0/in latch_0/tg_1/out 0
129 C40 latch_0/in d 0.00777f
130 C41 q vss 0.26331f
131 C42 d vss 0.47148f
132 C43 latch_1/tg_1/in vss 0.46607f
133 C44 invx1_2/in vss 1.13674f
134 C45 vdd vss 5.57422f
135 C46 latch_1/tg_1/out vss 0.70406f
136 C47 c vss 2.18871f
137 C48 latch_1/en vss 2.1965f
138 C49 latch_0/tg_1/in vss 0.46291f
139 C50 latch_1/in vss 0.86886f
140 C51 latch_0/tg_1/out vss 0.70406f
141 C52 latch_0/in vss 0.37981f
142 .ends
```

3.6 Verification

The DRC and LVS results are clean (Figure 12).

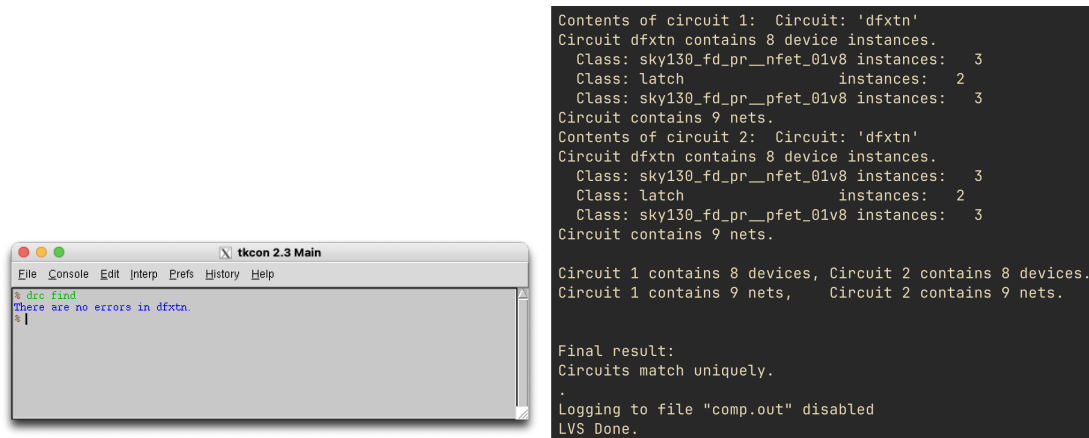
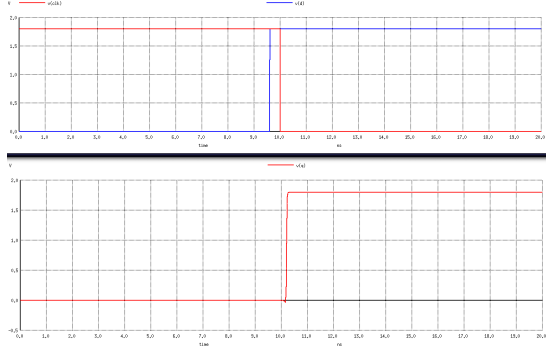


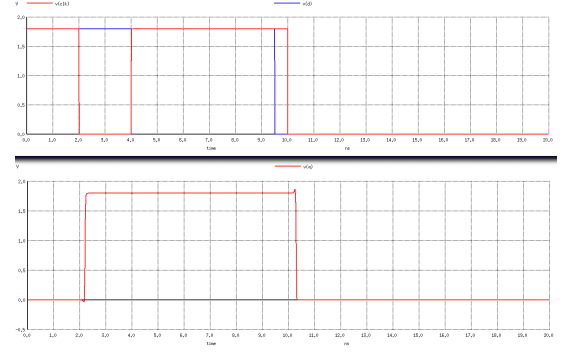
Figure 12: DRC and LVS clean results for the DFF cell

3.7 Waveforms

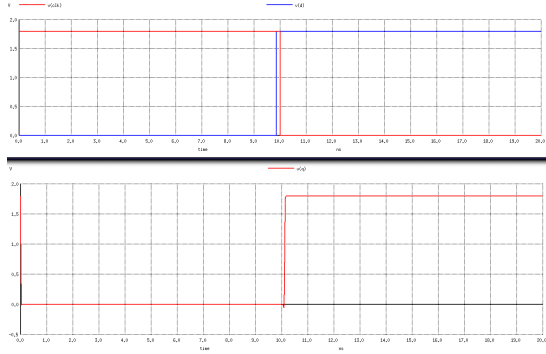
Waveforms from simulations showing input, output, and clock signals for rise and fall transitions, propagation delay, set-up time, and hold time are shown below. Each timing metric has separate plots for rise and fall transitions.



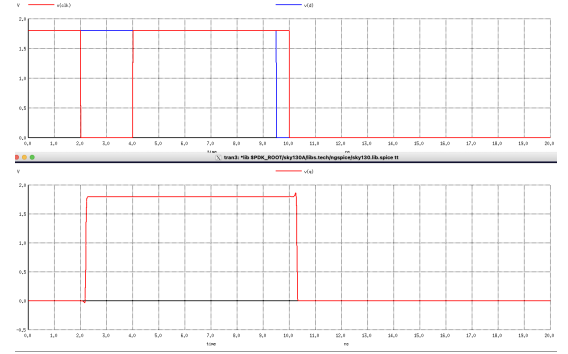
(a) Propagation delay rise (t_{cq})



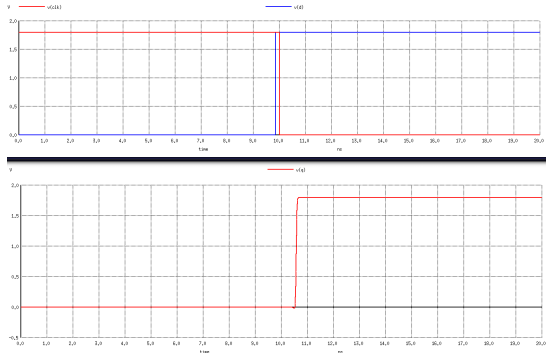
(b) Propagation delay fall (t_{cq})



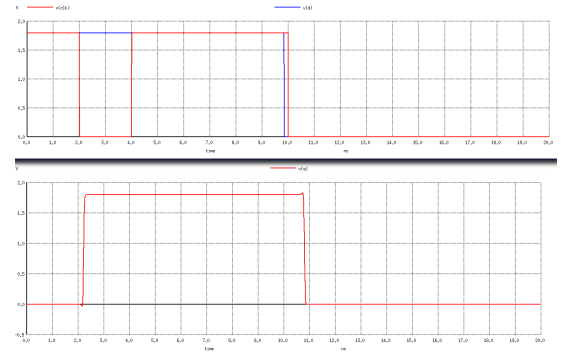
(c) Rise transition time (t_{rise})



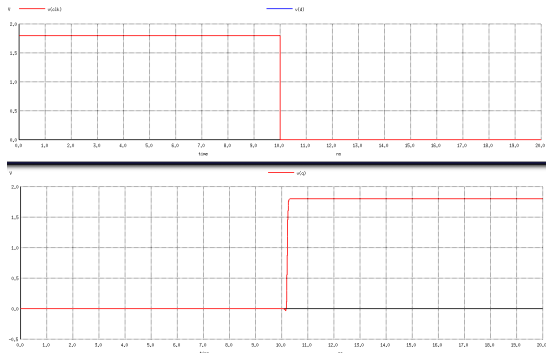
(d) Fall transition time (t_{fall})



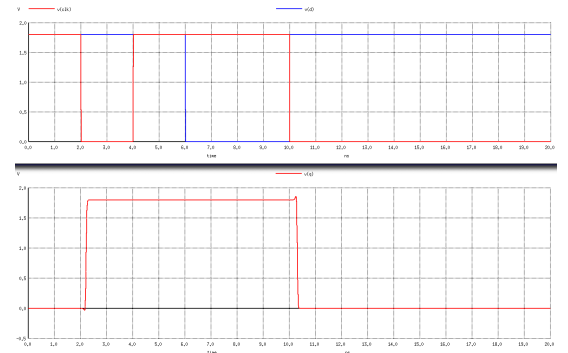
(e) Set-up time rise (t_{setup})



(f) Set-up time fall (t_{setup})



(g) Hold time rise (t_{hold})



(h) Hold time fall (t_{hold})

Figure 13: Waveforms of the DFF cell for rise and fall transitions and timing parameters

3.8 Timing and Power Characterisation

3.8.1 Input Pin Capacitances

Input Pins	Rise Cap (fF)	Fall Cap (fF)	Average Cap (fF)
D	4.01855	4.01825	4.0184
CLK	20.9715	20.9719	20.9717

Table 31: Input pin capacitances

3.8.2 Set-up Time

	10 ps	1000 ps
10 ps	0.152	0.333
1000 ps	0.243	0.409

Table 32: Rise Constraint (in ns) [Input slew vs CLK slew]

	10 ps	1000 ps
10 ps	0.158	0.355
1000 ps	0	0.17

Table 33: Fall Constraint (in ns) [Input slew vs CLK slew]

3.8.3 Hold Time

	10 ps	1000 ps
10 ps	0	0
1000 ps	0.034	0

Table 34: Rise Constraint (in ns) [Input slew vs CLK slew]

	10 ps	1000 ps
10 ps	0	0
1000 ps	0	0

Table 35: Fall Constraint (in ns) [Input slew vs CLK slew]

3.8.4 Transition Times (Q output, CLK slew = 10 ps)

	10 ps	100 ps	1000 ps
0.5 fF	0.018	0.018	0.018
10 fF	0.068	0.068	0.068
100 fF	0.596	0.596	0.596

Table 36: Output Rise Transition (in ns) [Input slew vs output capacitance]

	10 ps	100 ps	1000 ps
0.5 fF	0.027	0.027	0.027
10 fF	0.083	0.083	0.083
100 fF	0.644	0.644	0.644

Table 37: Output Fall Transition (in ns) [Input slew vs output capacitance]

3.8.5 CLK-to-Q Delay Time

	10 ps	100 ps	1000 ps
0.5 fF	0.196	0.196	0.196
10 fF	0.244	0.244	0.244
100 fF	0.596	0.596	0.596

Table 38: Cell Rise Delay (in ns) [Input slew vs output capacitance]

	10 ps	100 ps	1000 ps
0.5 fF	0.288	0.288	0.288
10 fF	0.346	0.346	0.346
100 fF	0.785	0.785	0.785

Table 39: Cell Fall Delay (in ns) [Input slew vs output capacitance]

3.8.6 Static Power (all input combinations)

Condition (CLK, D)	Power (nW)
00	4.226
01	5.252
10	0.429
11	0.612

Table 40: Static power for all input combinations of CLK and D

3.8.7 Dynamic Power (CLK slew = 10 ps)

	10 ps	100 ps	1000 ps
0.5 fF	37195.4	37038.6	37098.6
10 fF	44971.4	44814.6	44870.9
100 fF	117642	117485	117541

Table 41: Rise Power (in nW) [Input slew vs output capacitance]

	10 ps	100 ps	1000 ps
0.5 fF	42921.7	42821.2	42862.8
10 fF	43065.5	42965.1	43007.3
100 fF	43197.4	43097	43137.7

Table 42: Fall Power (in nW) [Input slew vs output capacitance]

3.9 Verilog

3.9.1 dfxtn Verilog Code

The Verilog code for the dfxtn cell used in simulations is shown below:

Listing 5: dfxtn Verilog code

```

1  `timescale 1ns/1ps
2
3  module dff(
4      input D,
5      input clk,
6      output reg Q
7  );
8      always @(negedge clk) begin
9          Q <= D;
10     end
11 endmodule

```

3.9.2 Testbench

The Verilog testbench used to verify the dfxtn functionality is shown below:

Listing 6: dfxtn Testbench code

```

1  `timescale 1ns/1ps
2
3  module tb;
4      reg D;
5      reg clk;
6      wire Q;
7
8      dff dut(.D(D), .clk(clk), .Q(Q));
9
10     initial clk = 0;
11     always #5 clk = ~clk;
12
13     initial begin
14         $dumpfile("dump.vcd");
15         $dumpvars(0, tb);
16         D = 0;
17         #7 D = 1;
18         #8 D = 0;
19         #12 D = 1;
20         #3 D = 0;

```

```

21         #20 D = 1;
22         #30 $finish;
23     end
24 endmodule

```

3.9.3 Simulation Waveform

The waveform from the Verilog simulation showing the input, output, and clock signals is shown in Figure 14.

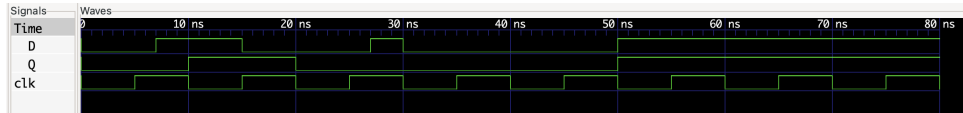


Figure 14: Verilog simulation waveform of the dfxtn cell

4 Work Distribution

Team Members	Contributions
Bhuvansh	Designed the layout, created the complete .lib file for DFXTN, and prepared the report.
Saarthak	Calculated characteristics for the combinational circuits, created Verilog file for the DFXTN, and prepared the report.
Hardik	Designed the layout for the combinational circuit and calculated the power characteristics for the inverter.
Shambhavi	Designed the layout, created the Verilog file, and calculated the timing characteristics for the combinational circuit.

Table 43: Distribution of work among team members during the project